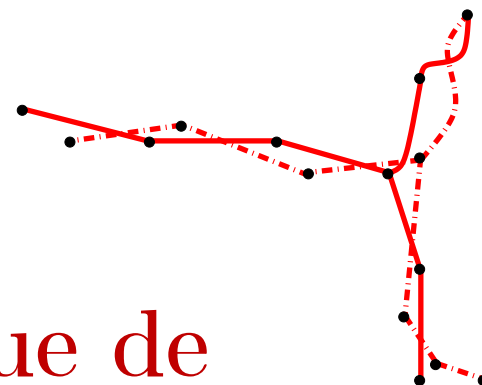
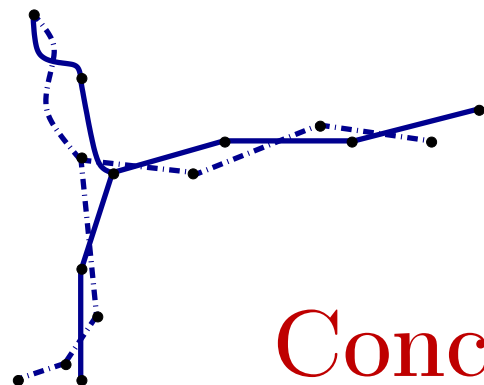




Conception et Pratique de l'Algorithmique

<http://www-npa.lip6.fr/~buixuan/cpa2023>

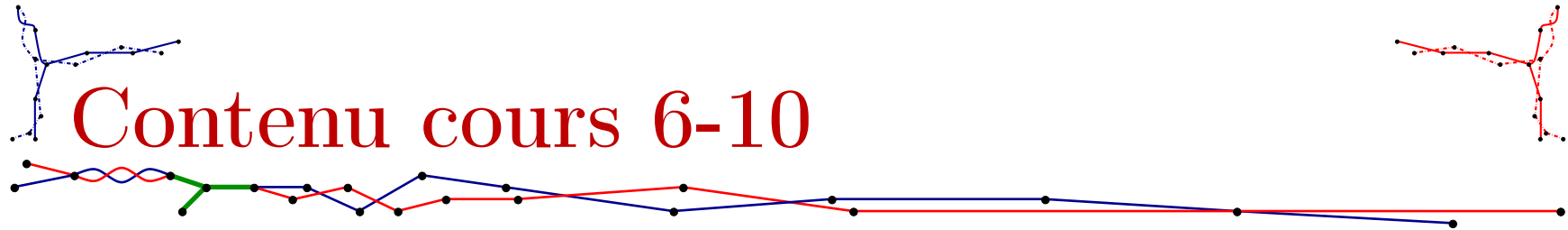
Binh-Minh Bui-Xuan



**SORBONNE
UNIVERSITÉ**
CRÉATEURS DE FUTURS
DEPUIS 1257

PARIS, Mars 2024

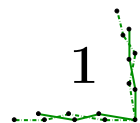
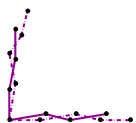


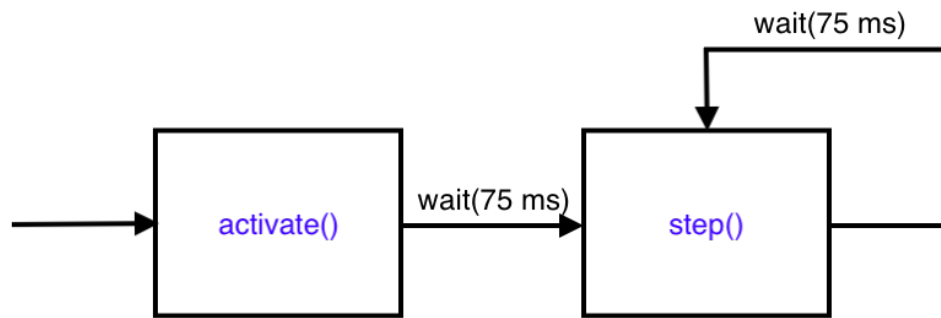
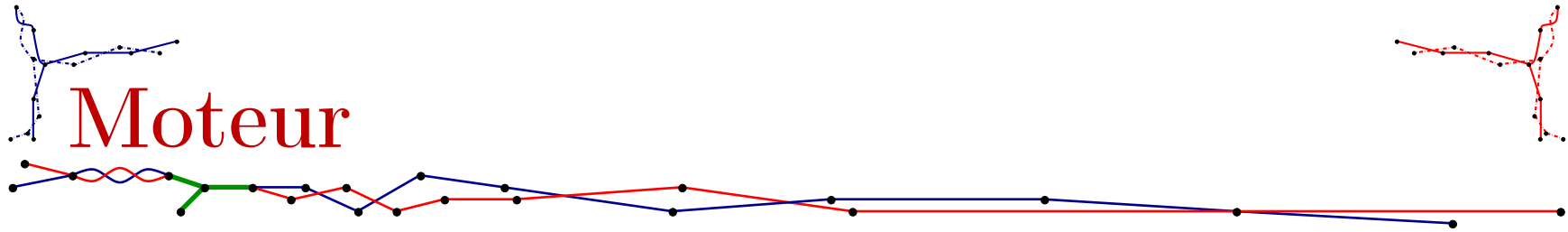


Contenu cours 6-10

SIMULATEUR :

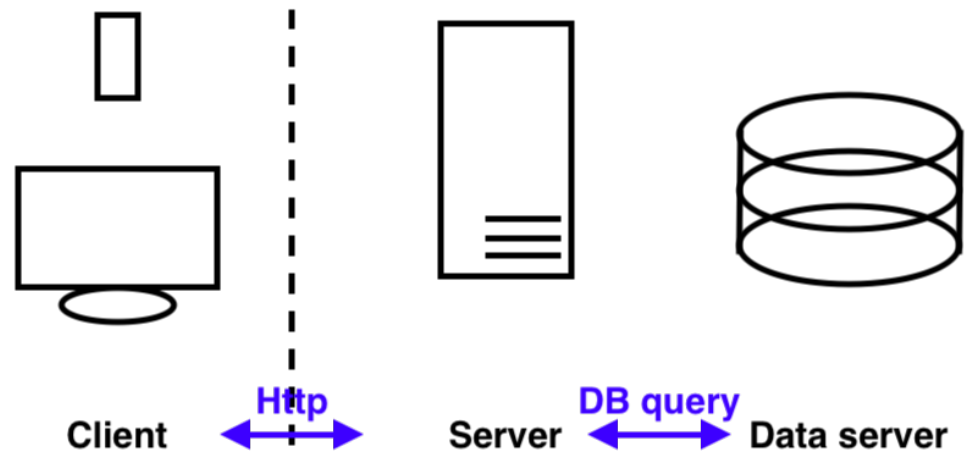
- logiciels CAD, visites virtuelles, jeux vidéo
- animation, effets spéciaux
- programmation embarquée
- en MU4IN505 : projet CPA





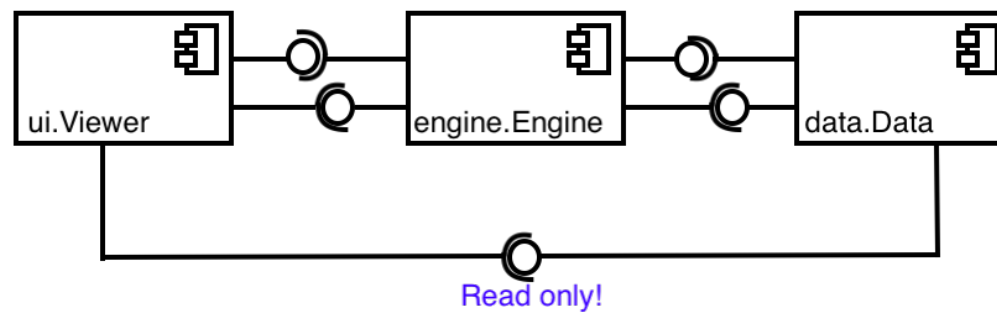
```
delta = 75ms;  
activate();  
wait(delta);  
while (true) {  
    step();  
    wait(delta);  
}
```

MVC online : 3 couches



- couche client $\approx$ partie Vue
- couche logique serveur $\approx$ partie Contrôleur
- couche serveur de DB $\approx$ partie Modèle

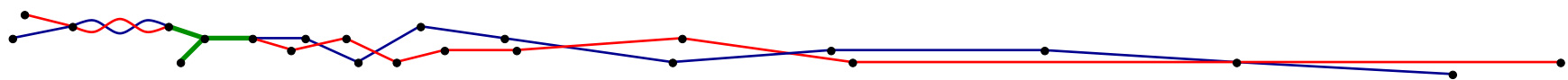
MVC desktop



- moteur de rendu graphique $\approx$ partie Vue
- moteur de jeu $\approx$ partie Contrôleur
- effets de bord, synchronisation $\approx$ partie Modèle



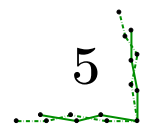
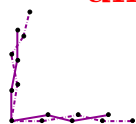
Exercice 1 : console sans interaction



```
void step() {  
    load_data();  
    compute();  
    save_data();  
  
    currentStep = load_data();  
    print(currentStep);  
    currentStep++;  
    save_data(currentStep);  
  
    if (!currentStep%10)  
        print(user_data);  
}
```

```
run:  
[echo] ===== Run Object Main =====  
[java] Game step #0: checked.  
[java] Heroes position: 1.0, 0.0  
[java] Game step #1: checked.  
[java] Game step #2: checked.  
[java] Game step #3: checked.  
[java] Game step #4: checked.  
[java] Game step #5: checked.  
[java] Game step #6: checked.  
[java] Game step #7: checked.  
[java] Game step #8: checked.  
[java] Game step #9: checked.  
[java] Heroes position: 8.0, -1.0  
[java] Game step #10: checked.  
[java] Game step #11: checked.
```

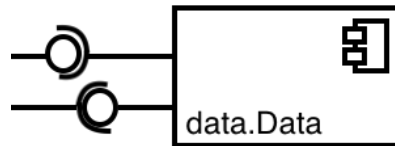
- affichage sur le console. Ex : snake, nethack, rogue-like, ...
- **un moteur** vs. deux moteurs? *I sync, therefore I am?*



Exercice 2 : windowed sans interaction

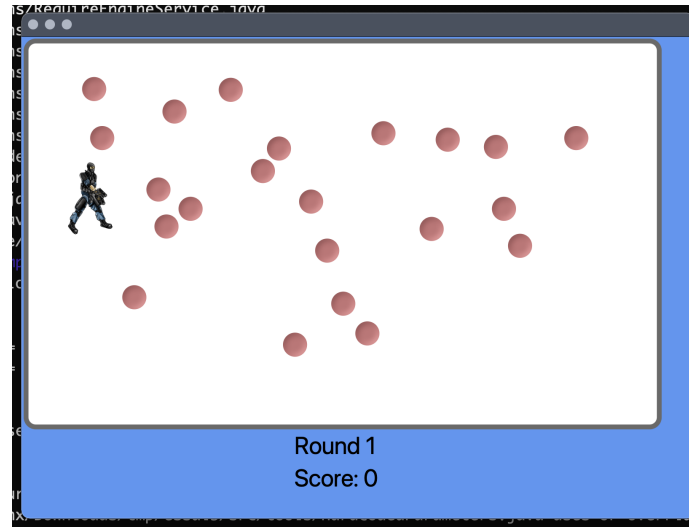
```
void step() {  
    load_data();  
    compute();  
    save_data();  
}
```

```
Image getPanel() {  
    load_data();  
    compute_image();  
    NEVER_save_data();  
}
```



- deux moteurs : rendu (Vue) + jeu (Contrôleur)
- synchronisation par mémoire partagée (Modèle)

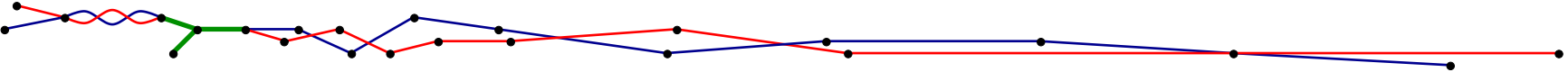
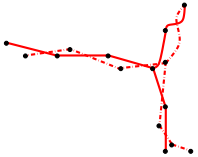
Exercice 3 : GUI responsive



- trois moteurs : commandes (Vue) + rendu (Vue) + jeu (Contrôleur)
- synchronisation par mémoire partagée (Modèle)



Gestion de commandes utilisateur

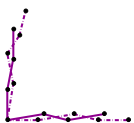


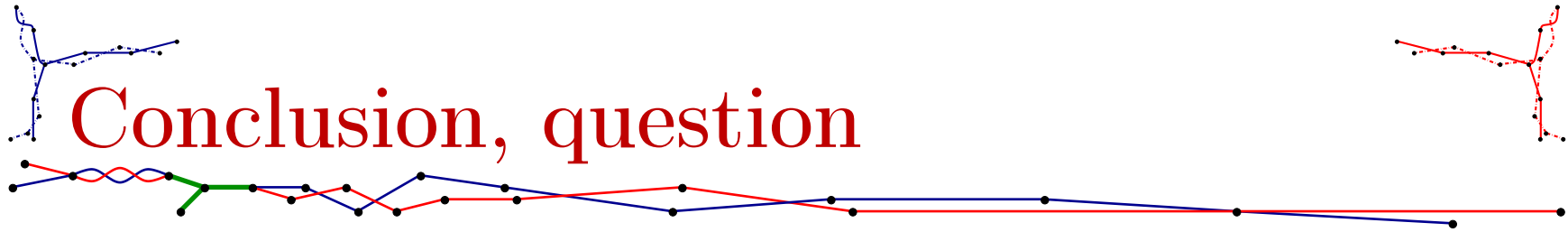
BITMAP :

- touche clavier $\longrightarrow$ position
- UX du 20e siècle

PHYSIQUE :

- 2e loi Newton : force $\approx$ accélération
- touche clavier $\longrightarrow$ vitesse
- UX de simulation naturelle





CONCLUSION :

- architecture MVC
- moteur :
 - de jeu
 - de rendu graphique
 - d'interaction utilisateur
- simulation naturelle

QUESTION :

- implantation ? (voir TME)

